

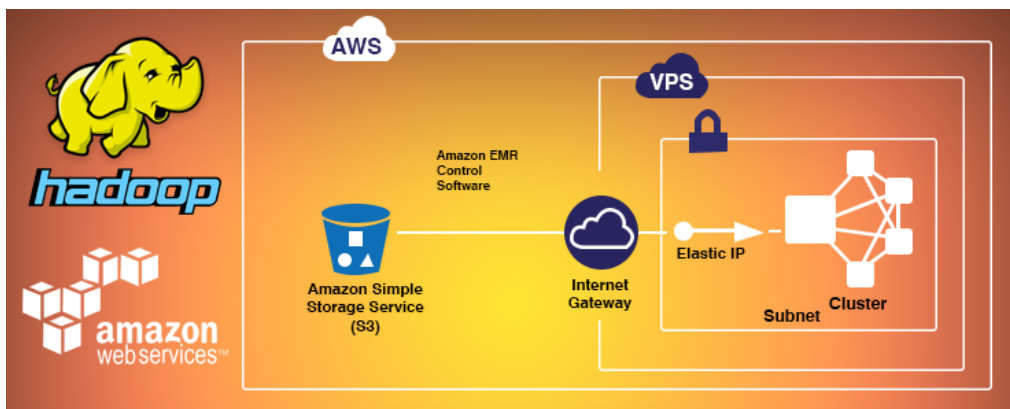
👤 [Abhishek \(https://www.eduonix.com/blog/author/rony/\)](https://www.eduonix.com/blog/author/rony/)

📅 July 9, 2015, 9 months ago

💬 2 (<https://www.eduonix.com/blog/bigdata-and-hadoop/a-step-by-step-guide-to-install-hadoop-cluster-on-amazon-ec2/#comments>)

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(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/Setting-Hadoop-Cluster-on-Amazon-EC2-Instance.png>)



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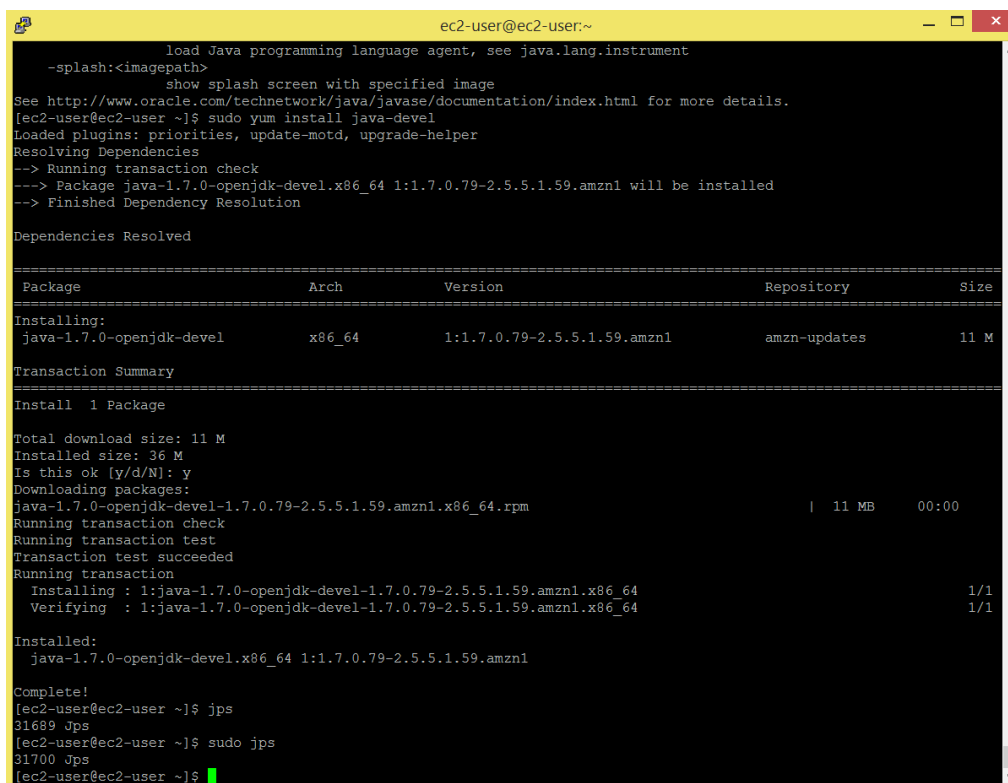
This is a step by step guide to install a Hadoop cluster on Amazon EC2. I have my AWS EC2 instance **ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com** ready on which I will install and configure Hadoop, java 1.7 is already installed.

In case java is not installed on you AWS EC2 instance, use below commands:

Command: `sudo yum install java-1.7.0-openjdk`

Command: `sudo yum install java-devel`

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/110.png>)



```
ec2-user@ec2-user:~$ sudo yum install java-devel
load Java programming language agent, see java.lang.instrument
--splash:<imagepath>
  show splash screen with specified image
See http://www.oracle.com/technetwork/java/javase/documentation/index.html for more details.
[ec2-user@ec2-user ~]$ sudo yum install java-devel
Loaded plugins: priorities, update-motd, upgrade-helper
Resolving Dependencies
--> Running transaction check
--> Package java-1.7.0-openjdk-devel.x86_64 1:1.7.0.79-2.5.5.1.59.amzn1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package                               Arch      Version                               Repository      Size
=====
Installing:
java-1.7.0-openjdk-devel              x86_64    1:1.7.0.79-2.5.5.1.59.amzn1         amzn-updates    11 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 11 M
Installed size: 36 M
Is this ok [y/d/N]: y
Downloading packages:
java-1.7.0-openjdk-devel-1.7.0.79-2.5.5.1.59.amzn1.x86_64.rpm | 11 MB  00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : 1:java-1.7.0-openjdk-devel-1.7.0.79-2.5.5.1.59.amzn1.x86_64      1/1
  Verifying   : 1:java-1.7.0-openjdk-devel-1.7.0.79-2.5.5.1.59.amzn1.x86_64      1/1
Installed:
  java-1.7.0-openjdk-devel.x86_64 1:1.7.0.79-2.5.5.1.59.amzn1

Complete!
[ec2-user@ec2-user ~]$ jps
31689 Jps
[ec2-user@ec2-user ~]$ sudo jps
31700 Jps
[ec2-user@ec2-user ~]$
```

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/110.png>)

I am installing hadoop-2.6.0 on the cluster. Below command will download hadoop-2.6.0 package.

Command: `wget`

<http://apache.mirrors.lucidnetworks.net/hadoop/common/had>

2.6.0/hadoop-2.6.0.tar.gz

```
ec2-user@ip-172-31-26-122:~  
[ec2-user@ip-172-31-26-122 ~]$ wget http://apache.mirrors.lucidnetworks.net/hadoop/common/hadoop-2.6.0/hadoop-2.6.0.tar.gz
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/25.png>

```
ec2-user@ip-172-31-26-122:~  
[ec2-user@ip-172-31-26-122 ~]$ wget http://apache.mirrors.lucidnetworks.net/hadoop/common/hadoop-2.6.0/hadoop-2.6.0.tar.gz  
--2015-06-30 16:12:58-- http://apache.mirrors.lucidnetworks.net/hadoop/common/hadoop-2.6.0/hadoop-2.6.0.tar.gz  
Resolving apache.mirrors.lucidnetworks.net (apache.mirrors.lucidnetworks.net)... 108.166.161.136  
Connecting to apache.mirrors.lucidnetworks.net (apache.mirrors.lucidnetworks.net)|108.166.161.136|:80... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 195257604 (186M) [application/x-gzip]  
Saving to: 'hadoop-2.6.0.tar.gz'  
  
hadoop-2.6.0.tar.gz          90%[=====>] 168.04M  474KB/s  eta 60s
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/34.png>

Check if the package got downloaded.

Command: ls

```
ec2-user@ip-172-31-26-122:~  
[ec2-user@ip-172-31-26-122 ~]$ ls  
hadoop-2.6.0.tar.gz  
[ec2-user@ip-172-31-26-122 ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/43.png>

Untar the file.

Command: tar -xvf hadoop-2.6.0.tar.gz

```
ec2-user@ip-172-31-26-122:~  
[ec2-user@ip-172-31-26-122 ~]$ tar -xvf hadoop-2.6.0.tar.gz
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/52.png>

```
ec2-user@ip-172-31-26-122:~
hadoop-2.6.0/share/hadoop/mapreduce/lib/xz-1.0.jar
hadoop-2.6.0/share/hadoop/mapreduce/lib/avro-1.7.4.jar
hadoop-2.6.0/share/hadoop/mapreduce/lib/log4j-1.2.17.jar
hadoop-2.6.0/share/hadoop/mapreduce/lib/guice-servlet-3.0.jar
hadoop-2.6.0/share/hadoop/mapreduce/hadoop-mapreduce-examples-2.6.0.jar
hadoop-2.6.0/share/hadoop/mapreduce/hadoop-mapreduce-client-hs-plugins-2.6.0.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-core-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-examples-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-jobclient-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-core-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-hs-plugins-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-shuffle-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-app-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-app-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-common-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-shuffle-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-examples-2.6.0-test-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-hs-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-hs-plugins-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-jobclient-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-common-2.6.0-sources.jar
hadoop-2.6.0/share/hadoop/mapreduce/sources/hadoop-mapreduce-client-hs-2.6.0-test-sources.jar
hadoop-2.6.0/LICENSE.txt
hadoop-2.6.0/README.txt
hadoop-2.6.0/bin/
hadoop-2.6.0/bin/hdfs.cmd
hadoop-2.6.0/bin/test-container-executor
hadoop-2.6.0/bin/container-executor
hadoop-2.6.0/bin/hadoop.cmd
hadoop-2.6.0/bin/rcc
hadoop-2.6.0/bin/hdfs
hadoop-2.6.0/bin/mapred
hadoop-2.6.0/bin/hadoop
hadoop-2.6.0/bin/yarn.cmd
hadoop-2.6.0/bin/mapred.cmd
hadoop-2.6.0/bin/yarn
hadoop-2.6.0/include/
hadoop-2.6.0/include/TemplateFactory.hh
hadoop-2.6.0/include/StringUtils.hh
hadoop-2.6.0/include/hdfs.h
hadoop-2.6.0/include/Pipes.hh
hadoop-2.6.0/include/SerialUtils.hh
[ec2-user@ip-172-31-26-122 ~]$ ls
hadoop-2.6.0  hadoop-2.6.0.tar.gz
[ec2-user@ip-172-31-26-122 ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/63.png>

Make hostname as ec2-user

Command: sudo hostname ec2-user

Command: hostname

```
ec2-user@ip-172-31-26-122:~
[ec2-user@ip-172-31-26-122 ~]$ sudo hostname ec2-user
[ec2-user@ip-172-31-26-122 ~]$ hostname
ec2-user
[ec2-user@ip-172-31-26-122 ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/73.png>

Below command will give you ip address, for me its 172.31.26.122

Command: ifconfig

```
ec2-user@ip-172-31-26-122:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 06:ED:BE:41:7D:7D
          inet addr:172.31.26.122  Bcast:172.31.31.255  Mask:255.255.240.0
          inet6 addr: fe80::4ed:beff:fe41:7d7d/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:9001  Metric:1
          RX packets:271138 errors:0 dropped:0 overruns:0 frame:0
          TX packets:109276 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:366530915 (349.5 MiB)  TX bytes:10016793 (9.5 MiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 b)  TX bytes:0 (0.0 b)

ec2-user@ip-172-31-26-122 ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/83.png>

Edit /etc/hosts file.

Command: sudo vi /etc/hosts

```
ec2-user@ec2-user:~$ sudo vi /etc/hosts
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/93.png>

Put ip address and hostname as below, save the file and close it.

```
ec2-user@ec2-user:~$ sudo vi /etc/hosts
172.31.26.122    localhost    ec2-user
~
~
~
~
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/103.png>

The '**ssh-agent**' is a background program that handles passwords for SSH private keys.

The '**ssh-add**' command prompts the user for a private key password and adds it to the list maintained by ssh-agent. Once you add a password to ssh-agent, you will not be asked to provide the key when using SSH or SCP to connect to hosts with your public key.

You will get **.pem** file from your amazon instance settings, copy it on amazon cluster, I have copied **hadoop.pem** file on my amazon ec2 cluster.

Protect key files to avoid any accidental or intentional corruption.

Command: `chmod 644 .ssh/authorized_keys`

Command: `chmod 400 hadoop.pem`

Start ssh-agent

Note: Make sure you use the backquote

(```), located under the tilde (`~`), rather than the single quote (`'`)

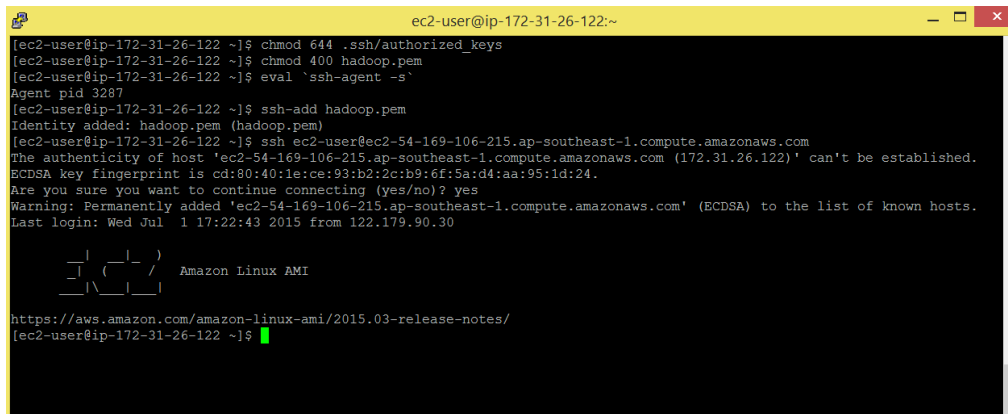
Command: `eval `ssh-agent -s``

Add the secure identity to SSH Agent Key repository

Command: `ssh-add hadoop.pem`

Command: `ssh ec2-user@ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com`

You will be able to login without password.



```
ec2-user@ip-172-31-26-122:~$ chmod 644 .ssh/authorized_keys
ec2-user@ip-172-31-26-122:~$ chmod 400 hadoop.pem
ec2-user@ip-172-31-26-122:~$ eval `ssh-agent -s`
Agent pid 3287
ec2-user@ip-172-31-26-122:~$ ssh-add hadoop.pem
Identity added: hadoop.pem (hadoop.pem)
ec2-user@ip-172-31-26-122:~$ ssh ec2-user@ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com
The authenticity of host 'ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com (172.31.26.122)' can't be established.
ECDSA key fingerprint is cd:80:40:1e:ce:93:b2:2c:b9:6f:5a:d4:aa:95:1d:24.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'ec2-54-169-106-215.ap-southeast-1.compute.amazonaws.com' (ECDSA) to the list of known hosts.
Last login: Wed Jul 1 17:22:43 2015 from 122.179.90.30

 _ _ _ _ _
| | | | |
|_|_|_|_|_| Amazon Linux AMI

https://aws.amazon.com/amazon-linux-ami/2015.03-release-notes/
ec2-user@ip-172-31-26-122:~$
```

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/113.png>)

Come out of the login.

Command: `exit`

Now we will add hadoop and java environment variables in `.bashrc` file.

Command: `sudo vi .bashrc`

```
ec2-user@ec2-user:~  
[ec2-user@ec2-user ~]$ sudo vi .bashrc
```

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/123.png>)

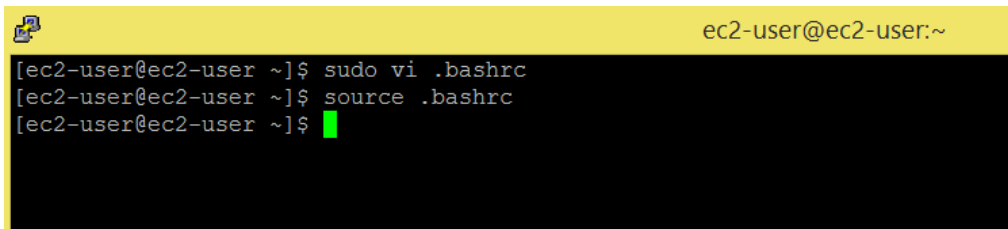
```
1 # Set Hadoop-related environment variables
2
3 export HADOOP_HOME=$HOME/hadoop-2.6.0
4
5 export HADOOP_CONF_DIR=$HOME/hadoop2.6.0/etc/hadoop
6
7 export HADOOP_MAPRED_HOME=$HOME/hadoop-2.6.0
8
9 export HADOOP_COMMON_HOME=$HOME/hadoop-2.6.0
10
11 export HADOOP_HDFS_HOME=$HOME/hadoop-2.6.0
12
13 export YARN_HOME=$HOME/hadoop-2.6.0
14
15 &nbsp;
16
17 # Set JAVA_HOME
18
19 export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_64
20
21 export PATH=$PATH:$JAVA_HOME/bin
22
23 &nbsp;
24
25 # Add Hadoop bin/ directory to PATH
26
27 export PATH=$PATH:$HOME/hadoop-2.6.0/bin
```

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/132.png>)

```
ec2-user@ec2-user:~  
# .bashrc  
  
# Source global definitions  
if [ -f /etc/bashrc ]; then  
    . /etc/bashrc  
fi  
  
# User specific aliases and functions  
  
# Set Hadoop-related environment variables  
  
export HADOOP_HOME=$HOME/hadoop-2.6.0  
export HADOOP_CONF_DIR=$HOME/hadoop2.6.0/etc/hadoop  
export HADOOP_MAPRED_HOME=$HOME/hadoop-2.6.0  
export HADOOP_COMMON_HOME=$HOME/hadoop-2.6.0  
export HADOOP_HDFS_HOME=$HOME/hadoop-2.6.0  
export YARN_HOME=$HOME/hadoop-2.6.0  
  
# Set JAVA_HOME  
  
export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_64/jre  
export PATH=$PATH:$JAVA_HOME/bin  
  
#Add adoop bin/ directory to PATH  
  
export PATH=$PATH:$HOME/hadoop-2.6.0/bin  
~  
~
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/132.png>

Command: source .bashrc

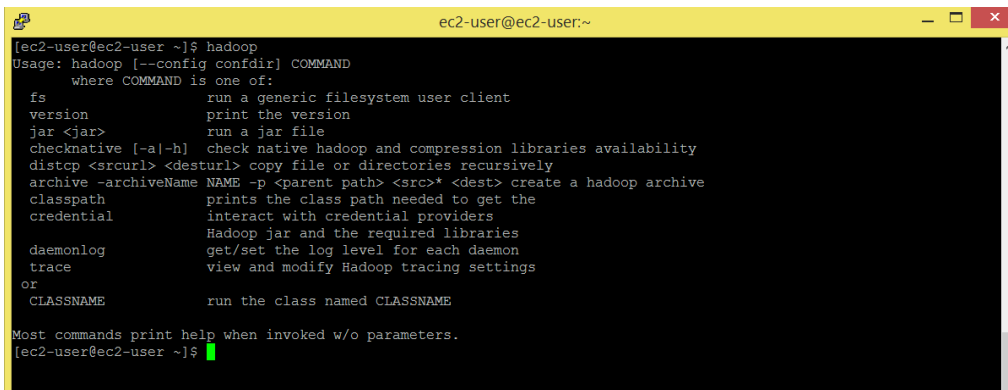


```
ec2-user@ec2-user:~  
[ec2-user@ec2-user ~]$ sudo vi .bashrc  
[ec2-user@ec2-user ~]$ source .bashrc  
[ec2-user@ec2-user ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/142.png>

Enter hadoop command and check if you get a set of options.

Command: hadoop

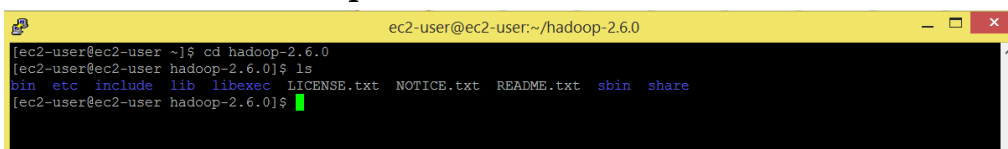


```
ec2-user@ec2-user:~  
[ec2-user@ec2-user ~]$ hadoop  
Usage: hadoop [--config confdir] COMMAND  
    where COMMAND is one of:  
    fs                run a generic filesystem user client  
    version            print the version  
    jar <jar>         run a jar file  
    checknative [-a|-h] check native hadoop and compression libraries availability  
    distcp <srcurl> <desturl> copy file or directories recursively  
    archive -archiveName NAME -p <parent path> <src>* <dest> create a hadoop archive  
    classpath          prints the class path needed to get the  
    credential         interact with credential providers  
    Hadoop jar and the required libraries  
    daemonlog         get/set the log level for each daemon  
    trace             view and modify Hadoop tracing settings  
    or  
    CLASSNAME         run the class named CLASSNAME  
  
Most commands print help when invoked w/o parameters.  
[ec2-user@ec2-user ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/151.png>

check out the directories present in hadoop-2.6.0

Command: cd hadoop-2.6.0



```
ec2-user@ec2-user:~/hadoop-2.6.0  
[ec2-user@ec2-user ~]$ cd hadoop-2.6.0  
[ec2-user@ec2-user hadoop-2.6.0]$ ls  
bin  etc  include  lib  libexec  LICENSE.txt  NOTICE.txt  README.txt  sbin  share  
[ec2-user@ec2-user hadoop-2.6.0]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/161.png>

share directory contains all the jar files.

Command: cd share/hadoop


```
ec2-user@ec2-user:~/hadoop-2.6.0/share/hadoop
[ec2-user@ec2-user hadoop-2.6.0]$ cd share/hadoop/
[ec2-user@ec2-user hadoop]$ ls
common hdfs httpfs kms mapreduce tools yarn
[ec2-user@ec2-user hadoop]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/171.png>

sbin directory contains all the script files to run or stop hadoop daemons/cluster.

```
ec2-user@ec2-user:~/hadoop-2.6.0/sbin
[ec2-user@ec2-user hadoop]$ cd ..
[ec2-user@ec2-user etc]$ cd ..
[ec2-user@ec2-user hadoop-2.6.0]$ cd sbin/
[ec2-user@ec2-user sbin]$ ls
distribute-exclude.sh  kms.sh  start-balancer.sh  stop-all.cmd  stop-yarn.cmd
hadoop-daemon.sh      mr-jobhistory-daemon.sh  start-dfs.cmd  stop-all.sh  stop-yarn.sh
hadoop-daemons.sh    refresh-namenodes.sh     start-dfs.sh   stop-balancer.sh  yarn-daemon.sh
hdfs-config.cmd       slaves.sh                 start-secure-dns.sh  stop-dfs.cmd  yarn-daemons.sh
hdfs-config.sh        start-all.cmd            start-yarn.cmd  stop-dfs.sh
httpfs.sh              start-all.sh             start-yarn.sh   stop-secure-dns.sh
[ec2-user@ec2-user sbin]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/181.png>

etc directory contains all the configuration files. We will edit few configuration files. Go to etc/hadoop/ directory.

Command: cd ..

Command: cd ..

Command: cd etc/hadoop

Command: ls

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ cd ..
[ec2-user@ec2-user share]$ cd ..
[ec2-user@ec2-user hadoop-2.6.0]$ cd etc/hadoop/
[ec2-user@ec2-user hadoop]$ ls
capacity-scheduler.xml  hadoop-policy.xml  kms-log4j.properties  ssl-client.xml.example
configuration.xml       hdfs-site.xml      kms-site.xml           ssl-server.xml.example
container-executor.cfg  httpfs-env.sh      log4j.properties      yarn-env.cmd
core-site.xml           httpfs-log4j.properties  mapred-env.cmd        yarn-env.sh
hadoop-env.cmd          httpfs-signature.secret  mapred-env.sh         yarn-site.xml
hadoop-env.sh           httpfs-site.xml      mapred-queues.xml.template
hadoop-metrics2.properties  kms-acls.xml      mapred-site.xml.template
hadoop-metrics.properties  kms-env.sh        slaves
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/191.png>

Set java home in **hadoop-env.sh** file.

```
1 # Set JAVA_HOME
2
3 export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_64
```

Command: sudo vi hadoop-env.sh

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/201.png>

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ sudo vi hadoop-env.sh
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/201.png>

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
# Licensed to the Apache Software Foundation (ASF) under one
# or more contributor license agreements. See the NOTICE file
# distributed with this work for additional information
# regarding copyright ownership. The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
# "License"); you may not use this file except in compliance
# with the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
# Set Hadoop-specific environment variables here.
#
# The only required environment variable is JAVA_HOME. All others are
# optional. When running a distributed configuration it is best to
# set JAVA_HOME in this file, so that it is correctly defined on
# remote nodes.
#
# The java implementation to use.
export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.79.x86_64/jre
#
# The jsvc implementation to use. Jsvc is required to run secure datanodes
# that bind to privileged ports to provide authentication of data transfer
# protocol. Jsvc is not required if SASL is configured for authentication of
# data transfer protocol using non-privileged ports.
#export JSVC_HOME=${JSVC_HOME}
#
export HADOOP_CONF_DIR=${HADOOP_CONF_DIR:-"/etc/hadoop"}
#
# Extra Java CLASSPATH elements. Automatically insert capacity-scheduler.
for f in $(HADOOP_HOME/contrib/capacity-scheduler/*.jar); do
  if [ "$HADOOP_CLASSPATH" ]; then
    export HADOOP_CLASSPATH=$HADOOP_CLASSPATH:$f
  else
    export HADOOP_CLASSPATH=$f
  fi
done
-- INSERT --
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/212.png>

Edit core-site.xml file. This file contains the configuration settings for Hadoop Core such as I/O settings that are common to HDFS and MapReduce.

Command: vi core-site.xml

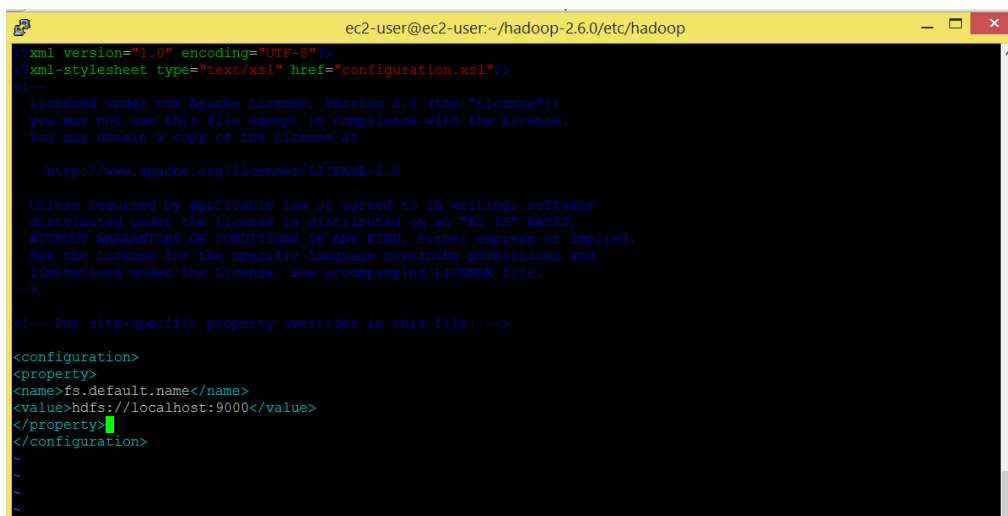
```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ vi core-site.xml
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/212.png>

[content/uploads/2015/07/221.png](http://www.eduonix.com/blog/wp-content/uploads/2015/07/221.png)

```
1 <strong>&lt;configuration&gt; </strong>
2
3 <strong>&lt;property&gt; </strong>
4
5 <strong>&lt;name&gt;fs.default.name&lt;/name&gt; </strong>
6
7 <strong>&lt;value&gt;hdfs://localhost:9000&lt;/value&gt; </strong>
8
9 <strong>&lt;/property&gt; </strong>
10
11 <strong>&lt;/configuration&gt;</strong>
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/231.png>



```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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distributed under the license is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->

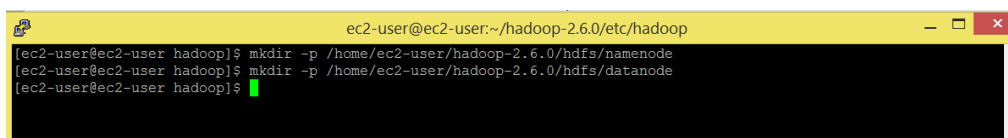
<configuration>
<property>
<name>fs.default.name</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/231.png>

Make namenode and datanode directory.

Command: mkdir -p /home/ec2-user/hadoop-2.6.0/hdfs/namenode

Command: mkdir -p /home/ec2-user/hadoop-2.6.0/hdfs/datanode

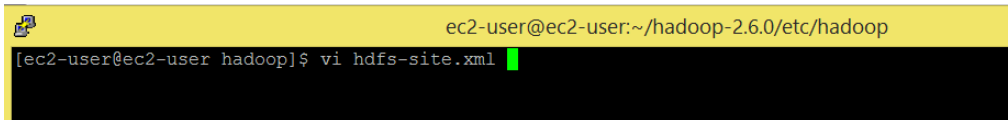


```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ mkdir -p /home/ec2-user/hadoop-2.6.0/hdfs/namenode
[ec2-user@ec2-user hadoop]$ mkdir -p /home/ec2-user/hadoop-2.6.0/hdfs/datanode
[ec2-user@ec2-user hadoop]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/241.png>

Edit `hdfs-site.xml`. This file contains the configuration settings for HDFS daemons; the Name Node, the secondary Name Node, and the data node.

Command: `vi hdfs-site.xml`



```
ec2-user@ec2-user: ~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ vi hdfs-site.xml
```

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/251.png>)

```
1 <configuration>
2
3 <property>
4
5 <name>dfs.replication</name>
6
7 <value>1</value>
8
9 </property>
10
11 <property>
12
13 <name>dfs.namenode.name.dir</name>
14
15 <value>/home/ec2-user/hadoop-2.6.0/hdfs/namenode</value>
16
17 </property>
18
19 <property>
20
21 <name>dfs.datanode.data.dir</name>
22
23 <value>>/home/ec2-user/hadoop-2.6.0/hdfs/datanode</value>
24
25 </property>
26
27 </configuration>
```

(<http://www.eduonix.com/blog/wp-content/uploads/2015/07/26.png>)

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
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WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>/home/ec2-user/hadoop-2.6.0/hdfs/namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/home/ec2-user/hadoop-2.6.0/hdfs/datanode</value>
</property>
</configuration>
~
~
~
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/26.png>

Edit mapred-site.xml. This file contains the configuration settings for MapReduce daemons.

Command: cp mapred-site.xml.template mapred-site.xml

Command: vi mapred-site.xml

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ cp mapred-site.xml.template mapred-site.xml
[ec2-user@ec2-user hadoop]$ vi mapred-site.xml
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/27.png>

```
1  <configuration>
2
3  <property>
4
5  <name>mapreduce.framework.name</name>
6
7  <value>yarn</value>
8
9  </property>
10
11 </configuration>
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/28.png>

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
</configuration>
~
~
~
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/28.png>

Edit yarn-site.xml. This file contains the configuration settings for YARN.

Command: vi yarn-site.xml

```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
[ec2-user@ec2-user hadoop]$ vi yarn-site.xml
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/29.png>

```
1 <configuration>
2 <property>
3   <name>yarn.nodemanager.aux-services</name> <value>mapreduce_shuffle</value>
4 </property>
5 <property>
6   <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
7   <value>org.apache.hadoop.mapred.ShuffleHandler</value>
8 </property>
9 </configuration>
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/30.png>



```
ec2-user@ec2-user:~/hadoop-2.6.0/etc/hadoop
<?xml version="1.0"?>
<!--
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WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Site specific YARN configuration properties -->
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name> <value>mapreduce_shuffle</value>
</property>
<property>
<name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
<value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
</configuration>
~
~
~
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/30.png>

Now we are ready to start the cluster. We will format the namenode.

Command: cd

Command: hadoop namenode -format

```
ec2-user@ec2-user:~
[ec2-user@ec2-user ~]$ cd
[ec2-user@ec2-user ~]$ hadoop namenode -format
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/311.png>

```
ec2-user@ec2-user:~  
2015-06-30 21:34:05,657 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(361)) - capacity = 2^20 = 1048576 entries  
2015-06-30 21:34:05,659 INFO [main] namenode.NameNode (FSDirectory.java:<init>(234)) - Caching file names occurring more than 10 times  
2015-06-30 21:34:05,667 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(354)) - Computing capacity for map cachedBlocks  
2015-06-30 21:34:05,667 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(355)) - VM type = 64-bit  
2015-06-30 21:34:05,668 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(356)) - 0.25% max memory 966.7 MB = 2.4 MB  
2015-06-30 21:34:05,668 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(361)) - capacity = 2^18 = 262144 entries  
2015-06-30 21:34:05,669 INFO [main] namenode.FSNamesystem (FSNamesystem.java:<init>(5663)) - dfs.namenode.safemode.threshold-pct = 0.9990000128746033  
2015-06-30 21:34:05,669 INFO [main] namenode.FSNamesystem (FSNamesystem.java:<init>(5664)) - dfs.namenode.safemode.min.datanodes = 0  
2015-06-30 21:34:05,669 INFO [main] namenode.FSNamesystem (FSNamesystem.java:<init>(5665)) - dfs.namenode.safemode.extension = 30000  
2015-06-30 21:34:05,670 INFO [main] namenode.FSNamesystem (FSNamesystem.java:initRetryCache(957)) - Retry cache on namenode is enabled  
2015-06-30 21:34:05,670 INFO [main] namenode.FSNamesystem (FSNamesystem.java:initRetryCache(965)) - Retry cache will use 0.03 of total heap and retry cache expiry time is 600000 millis  
2015-06-30 21:34:05,672 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(354)) - Computing capacity for map NameNodeRetryCache  
2015-06-30 21:34:05,673 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(355)) - VM type = 64-bit  
2015-06-30 21:34:05,673 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(356)) - 0.029999999329447746% max memory 966.7 MB = 297.0 KB  
2015-06-30 21:34:05,673 INFO [main] util.GSet (LightWeightGSet.java:computeCapacity(361)) - capacity = 2^15 = 32768 entries  
2015-06-30 21:34:05,686 INFO [main] namenode.NNConf (NNConf.java:<init>(62)) - ACLs enabled? false  
2015-06-30 21:34:05,686 INFO [main] namenode.NNConf (NNConf.java:<init>(66)) - XAttrs enabled? true  
2015-06-30 21:34:05,686 INFO [main] namenode.NNConf (NNConf.java:<init>(74)) - Maximum size of an xattr: 16384  
Re-format filesystem in Storage Directory /tmp/hadoop-ec2-user/dfs/name ? (Y or N) Y  
2015-06-30 21:34:16,035 INFO [main] namenode.FSImage (FSImage.java:format(145)) - Allocated new BlockPoolId: BP-1097253001-172.31.26.122-1435700055961  
2015-06-30 21:34:16,053 INFO [main] common.Storage (NNStorage.java:format(552)) - Storage directory /tmp/hadoop-ec2-user/dfs/name has been successfully formatted.  
2015-06-30 21:34:16,277 INFO [main] namenode.NNStorageRetentionManager (NNStorageRetentionManager.java:getImageTxIdToRetain(203)) - Going to retain 1 images with txid >= 0  
2015-06-30 21:34:16,282 INFO [main] util.ExitUtil (ExitUtil.java:terminate(124)) - Exiting with status 0  
2015-06-30 21:34:16,289 INFO [Thread-1] namenode.NameNode (StringUtils.java:run(659)) - SHUTDOWN_MSG:  
/*****  
SHUTDOWN_MSG: Shutting down NameNode at ec2-user/172.31.26.122  
*****/  
[ec2-user@ec2-user ~]$
```

<http://www.eduonix.com/blog/wp-content/uploads/2015/07/321.png>

After we format the namenode successfully, we will start all the hadoop daemons.

You are now all set to start the HDFS services i.e. Name Node, Secondary Name Node, and Data Node on your Hadoop Cluster.

Command: cd hadoop-2.6.0/sbin/

Command: ./start-dfs.sh

Start the YARN services i.e. ResourceManager and NodeManager

Command: ./start-yarn.sh

Now run jps command to check if the daemons are running.

Command: jps

Hadoop
cluster
on
Amazon
EC2&summary
is a
step by
step
guide
to
install
a
Hadoop
cluster
on
Amazon
EC2. I
have
my
AWS
EC2
instance
ec2-54-
169-
106-
215.ap-
southeast-
1.compute
ready
on
which I
will
install
and
configure
Hadoop,
java
1.7 is
already
installed.
In case
java is

not
installed
on you
AWS
EC2
instance.
)



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Myles Gartland · Professor at Rockhurst University

in your .bashrc should your "export HADOOP_CONF_DIR=\$HOME dash in it?

"export HADOOP_CONF_DIR=\$HOME/hadoop-2.6.0/etc/hadoop

Like · Reply · Nov 27, 2015 2:44pm



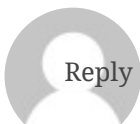
Naveen Mandadhi · Northern Illinois University

yes, it should! I struggled with it for so long until I figured

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Vijju

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[amazon-ec2/#comment-10982](https://www.eduonix.com/blog/bigdata-and-hadoop/a-step-by-step-guide-to-install-hadoop-cluster-on-amazon-ec2/#comment-10982))

Hello, I followed the steps above. But, am getting below error.

```
[ec2-user@ip-10-177-1-69 sbin]$ ./start-dfs.sh
Incorrect configuration: namenode address
dfs.namenode.servicerpc-address or
dfs.namenode.rpc-address is not configured.
Starting namenodes on []
Error: Cannot find configuration directory:
/home/ec2-user/hadoop2.6.0/etc/hadoop
Error: Cannot find configuration directory:
/home/ec2-user/hadoop2.6.0/etc/hadoop
Starting secondary namenodes [0.0.0.0]
Error: Cannot find configuration directory:
/home/ec2-user/hadoop2.6.0/etc/hadoop
```



Naveen

Reply (https://www.eduonix.com/blog/bigdata-and-hadoop/a-step-by-step-guide-to-install-hadoop-cluster-on-amazon-ec2/?replytocom=11447#respond)

(https://www.eduonix.com/blog/bigdata-and-

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I am getting the same error. Please help if you found solution.

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